DIP JOTI PAUL

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EDUCATION

Massachusetts Institute of Technology (MIT), Massachusetts, USA
Ph.D. Candidate, Electrical Engineering and Computer Science (EECS) 01/2021 – present
Advisor: Prof. Karl Berggren
Research topics: Photonics, Superconducting detectors and electronics, Nanofabrication

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

M.Sc., Electrical and Electronic Engineering (EEE)	CGPA: 4.0/4.0	08/2020
B.Sc., Electrical and Electronic Engineering (EEE)	Class rank: 3/194	03/2017

HONORS and ACHIEVEMENTS

- IEEE Council on Superconductivity (CSC) Graduate Fellowship (2026)
- 2025-2026 MathWorks Fellow, MIT School of Engineering
- MIT RLE Claude E. Shannon Fellowship (2025); featured in Weston Observer
- Honorable Mention, APS Division of Laser Science Student Poster Competition, CLEO 2022
- Dean's List for Academic Excellence, each undergraduate year (BUET)

SELECTED PUBLICATIONS

1. F. Incalza, M. Castellani, **D.J. Paul**, A. Simon, E. Batson, D. Mondin, O. Medeiros, and K.K. Berggren, "Fast-Recovery Epitaxial NbN Superconducting Nanowire Single-Photon Detectors with Saturated Efficiency at 1550 nm in Liquid Helium", Nano Letters (2026).
2. S. Koppell*, O. Bittencourt*, **D.J. Paul**, M. Baryakhtar, J. Huang, and K.K. Berggren, "Dark Matter Haloscope with a Disordered Dielectric Absorber", Physical Review D (2026).
3. **D.J. Paul**, T.X. Zhou, and K.K. Berggren, "Photolithography Compatible Three-Terminal Superconducting Switch for Driving CMOS Loads", Physical Review Applied (2025).
4. **D.J. Paul**, T.X. Zhou and K.K. Berggren, "Determination of Mid-Infrared Refractive Indices of Superconducting Thin Films Using Fourier Transform Infrared Spectroscopy", Applied Physics Letters (2025).
5. M.I. Tahmid, **D.J. Paul**, and M.Z. Baten, "Emergence and Tunability of Transmission Gap in the Strongly Disordered Regime of a Dielectric Random Scattering Medium", Optics Express, OSA Publishing (2021).
6. **D.J. Paul**, S. Hossain, and M.Z. Baten, "Disorder Induced Rotational-Symmetry Breaking to Control Directionality of Whispering Gallery Modes in Circularly Symmetric Nanowire Assembly", Optics Express, OSA Publishing (2019).
7. **D.J. Paul**, A.A. Mimi, A. Hazari, P. Bhattacharya, and M.Z. Baten, "Finite-Difference Time-Domain Analysis of the Tunability of Anderson Localization of Light in Self-Organized GaN Nanowire Arrays", Journal of Applied Physics, AIP Publishing (2019).

Under Review:

1. **D.J. Paul**, S. Koppell, G.G. Taylor, B. Korzh, S.R. Patel, A.D. Beyer, E.E. Wollman, M.D. Shaw, P. D. Keathley, and K.K. Berggren, "Enhanced Mid-Infrared Single-Photon Detection with Antenna-Coupled Superconducting Nanowires", under review at Nano Letters.

INVITED AND CONFERENCE TALKS

- Applied Superconductivity Conference, Pittsburgh, Pennsylvania — 09/2026
  ”Kinetic inductance and relaxation oscillation dynamics in thin-film $YBa_2Cu_3O_{7-x}$ nanowires”
- BREAD Collaboration Workshop, Harvard University — 01/2026
  ”Mid-infrared SNSPDs for Exploring Dark Matter” — Host: Prof. Stefan Knirck
- RLE Shannon Fellowship TED Talk, Weston Middle School — 12/2025
  ”Beyond What Our Eyes Can See” — Host: Jaye Goldstein
- Applied Superconductivity Conference, Honolulu, Hawaii — 10/2022
  ”Optimizing Infrared Sensitivity of Superconducting Nanowire Single-Photon Detectors”

RESEARCH GRANT WRITING EXPERIENCE

1. NSF EPMD Program with Prof. K. Berggren, Prof. D. Santavicca, and Prof. M. Warusawithana; wrote two thrusts on high-operating-temperature SNSPDs using YBCO thin films. [declined]
2. DOE QIS-Enabled High Energy Physics (QuantISED 2.0) with Prof. L. Winslow, Prof. K. Berggren, Prof. M. Baryakhtar, Prof. G. Rybka, and Prof. R. Maruyama; contributed to proposal development and wrote one thrust on mid-infrared SNSPD development for the proposed pathfinder experiments. [awarded in 12/2024; news coverage]
3. NASA APRA ROSES 2023 with B. Korzh and Prof. K. Berggren; wrote one thrust on antenna coupling to improve mid-infrared absorption efficiency in SNSPDs. [awarded in 06/2024]

PROFESSIONAL EXPERIENCE

- Research Assistant, Quantum Nanostructures and Nanofabrication Group, MIT
  Advsior: Prof. Karl Berggren — 01/2021 – present
  Experience with superconducting device measurements and SNSPD characterization; performed research to improve mid-infrared sensitivity and increase the operating temperature of SNSPDs.
- Research Assistant, Solid-state Electronics and Photonics Group, BUET
  Advsior: Prof. Md Zunaid Baten — 03/2017 – 06/2020
  Developed FDTD simulation framework to study light–matter interactions in nanoscale devices.
- Research Assistant, Nano Device Research Group, BUET
  Advsior: Prof. Quazi D. M. Khosru — 03/2016 – 02/2017
  Developed a self-consistent charge-transport simulation model for tunneling FET.

TEACHING AND MENTORSHIP EXPERIENCE

- Teaching Assistant, 6.2540: Nanotechnology: From Atoms to Systems, MIT EECS — Fall 2023
  Applied quantum mechanics and nanofabrication techniques (undergraduate).
- Lecturer, Bangladesh University of Engineering and Technology — 05/2017 – 08/2020
  Courses taught: Electronic Circuits, Digital Electronics, and VLSI.
- Undergraduate Research Mentor (UROP), MIT EECS — 2024–Present
  Mentored three MIT undergraduates on superconducting device simulations and cryogenic measurements.

TECHNICAL EXPERTISE

- Nanofabrication Process Development and Material Characterization: Nanolithography training in the MIT.nano Class-100 cleanroom; fabricated devices down to 15 nm linewidth.
- Programming and Data Analysis: MATLAB, Python, C++.
- Multiphysics and EM Simulation: COMSOL Multiphysics, Ansys Lumerical, MEEP FDTD.
- Circuit Design and Simulation: Cadence Virtuoso, Silvaco TCAD, Verilog, LTspice.